

# UrbanCart Marketing Campaign Performance Analytics

End-to-end Power BI marketing analytics project covering campaign performance, marketing funnel efficiency, channel effectiveness, customer segments, geography, and return on marketing investment.

|                   |                                                          |
|-------------------|----------------------------------------------------------|
| <b>Project</b>    | UrbanCart Marketing Campaign Performance Analytics       |
| <b>Period</b>     | 2024-2025                                                |
| <b>Technology</b> | Power BI • Power Query • DAX • Data Modelling            |
| <b>Dataset</b>    | Synthetic marketing dataset                              |
| <b>Campaigns</b>  | 25 campaign names in the final Campaign Performance view |
| <b>Author</b>     | Tejaswi Aluka                                            |

Portfolio / Learning Project

# Executive Summary

UrbanCart is a fictional retail business created to demonstrate an end-to-end Data Analyst and Business Intelligence workflow. The project uses Power BI to analyse marketing performance across campaigns, channels, customer segments, devices and geography, with an emphasis on turning marketing data into actionable business insights.

The full Campaign Performance view reports \$130.97M in marketing spend, \$241.36M in revenue, 1.84x ROAS, 2,168,828 conversions, \$60.39 CPA and 84.29% ROI. These figures represent the full campaign view and may differ from filtered dashboard pages.

## Business Problem

UrbanCart needs a consolidated view of marketing performance to understand:

- How much revenue is generated from marketing investment?
- How efficiently is marketing spend converted into revenue?
- How are customers progressing through the marketing funnel?
- Which campaigns generate stronger revenue, conversions and returns?
- Which channels and customer groups contribute to engagement and conversions?
- Where are the major funnel drop-offs?
- Which historical campaign patterns could inform future campaign testing?

## Business Objectives

- Measure overall marketing investment and revenue performance.
- Monitor ROAS, ROI, CPA, CPC, CTR and conversion performance.
- Analyse the marketing funnel from impressions through conversions.
- Compare campaign-level revenue, spend, conversions and efficiency.
- Analyse channel, customer segment, device and geographic performance.
- Identify funnel drop-offs and potential areas for optimisation.
- Distinguish customer acquisition performance from retention and reactivation.
- Provide an executive-friendly Power BI dashboard to support marketing decision-making.

## Dataset & Technology

The project uses a synthetic UrbanCart marketing dataset covering 2024–2025. Key fields include campaign, date, channel, customer, customer segment, age group, device, state, city, impressions, clicks, leads, conversions, spend and revenue.

The final Campaign Performance view contains 25 campaign names.

- Power BI
- DAX
- Power Query
- Data modelling
- Data quality validation

- Business intelligence
- Dashboard design
- GitHub

## Project Structure

```
UrbanCart-Marketing-Analytics/
■■■ README.md
■■■ .gitignore
■■■ UrbanCart Marketing Campaign Performance Report.pdf
■■■ data/
■■■ powerbi/
■ ■■■ UrbanCart_marketing_campaign_performance_dashboard.pbix
■■■ dax/
■■■ documentation/
■■■ screenshots/
```

## Data Quality & Cleaning

| Check                | Action / Finding                                                                           |
|----------------------|--------------------------------------------------------------------------------------------|
| Duplicate records    | 150 duplicates identified and removed                                                      |
| State values         | Unknown values investigated and corrected/derived using city information where appropriate |
| Missing Spend        | Missing values identified and investigated                                                 |
| Missing Revenue      | Missing values identified and investigated                                                 |
| Clicks > Impressions | Invalid records flagged for investigation                                                  |
| Conversions > Clicks | Invalid records flagged for investigation                                                  |
| Category consistency | Values standardised and mapping tables created                                             |

## Data Model

The Power BI model uses a fact-and-dimension approach.

### Fact table

**Fact\_Marketing** includes campaign, date, channel, geography, customer, segment, device, impressions, clicks, leads, conversions, spend and revenue.

### Dimension tables

- Dim\_Date
- Dim\_Campaign
- Dim\_Channel
- Dim\_Geography
- Channel Mapping
- Segment Mapping

## Key DAX Measures

| Measure           | Definition                                                |
|-------------------|-----------------------------------------------------------|
| Total Spend       | SUM(Fact_Marketing[Clean_Spend])                          |
| Total Revenue     | SUM(Fact_Marketing[Clean_Revenue])                        |
| Total Conversions | SUM(Fact_Marketing[Clean_Conversions])                    |
| ROAS              | DIVIDE([Total Revenue], [Total Spend], 0)                 |
| Conversion Rate % | DIVIDE([Total Conversions], [Total Clicks], 0)            |
| CTR %             | DIVIDE([Total Clicks], [Total Impressions], 0)            |
| CPC               | DIVIDE([Total Spend], [Total Clicks], 0)                  |
| CPA               | DIVIDE([Total Spend], [Total Conversions], 0)             |
| ROI               | DIVIDE([Total Revenue] - [Total Spend], [Total Spend], 0) |

# Business Questions & Key Findings

## Overall Performance

- How much are we spending on marketing?
- How much revenue is being generated?
- What is the overall ROAS and ROI?
- How many leads and conversions are being generated?
- Is marketing efficiency improving or declining over time?

## Funnel

- How many users move from impressions to clicks?
- How many clicks become leads?
- How many leads become conversions?
- Where are the largest funnel drop-offs?
- How is conversion rate changing over time?

## Campaigns

- Which campaigns generate the most revenue?
- Which campaigns generate the highest conversion volume?
- Which campaigns have stronger ROAS?
- Which campaigns have lower CPA?
- Are high-volume campaigns also efficient?

## Channels

- Which channels generate stronger engagement?
- Which channels convert more effectively?
- Which channels contribute the most conversions?
- How do owned channels compare with paid media?

## Customers

- How much conversion volume comes from new versus existing customers?
- Which customer segments contribute the most conversions?
- Should acquisition and retention performance be evaluated separately?

## Geography & Devices

- Which states and cities contribute the most revenue and conversions?
- Does device type affect conversion performance?
- Where are potential geographic opportunities for further investigation?

# Key Insights

## 1. UrbanCart is scaling revenue, but efficiency is broadly flat

UrbanCart generated \$241.36M revenue from \$130.97M marketing spend, delivering 1.84x ROAS. Revenue and spend have both almost doubled versus the previous period, while ROAS has moved only slightly from 1.86x to 1.84x.

**Business impact:** UrbanCart is successfully scaling, but growth is currently requiring a broadly proportional increase in marketing investment.

**Management implication:** The next growth target should focus on improving incremental return from existing spend, rather than relying purely on additional budget.

## 2. Payday Promotion delivers scale, but scale alone doesn't justify more budget

Payday Promotion generated approximately 90.8K conversions and \$10.17M revenue. It required \$5.54M in spend, producing 1.83x ROAS, close to the overall portfolio ROAS of 1.84x.

**Business impact:** Payday Promotion is an important volume driver, but its large conversion contribution does not necessarily mean superior efficiency.

**Management implication:** Before increasing investment, evaluate new-customer acquisition, average order value, gross margin, repeat purchase rate and customer lifetime value.

## 3. UrbanCart's biggest visible funnel opportunity is after lead generation

UrbanCart generates 33.12M leads, but only 2.17M conversions. The Lead → Conversion rate is approximately 6.55%, meaning 93.45% of leads do not currently progress to conversion.

**Business impact:** The dashboard suggests there may be more value in improving conversion of existing leads than simply generating additional traffic.

**Management implication:** Investigate lead quality, follow-up timing, offer relevance, landing-page experience, remarketing and checkout friction.

## 4. Email is a major conversion engine — but attribution needs to be separated from incrementality

Email has the highest visible CTR at 8.62% and the highest channel conversion rate at 1.81%. It also contributes approximately 1.04M conversions, nearly half of total reported conversions.

**Business impact:** UrbanCart's customer database appears to be a major commercial asset.

**Management implication:** Determine how much incremental revenue Email generates versus purchases that would have occurred anyway.

## 5. Existing customers are a major source of UrbanCart's conversion volume

Approximately 62% of conversions come from Returning, Loyal and At-Risk customers combined.

**Business impact:** UrbanCart is not purely an acquisition-led business.

**Management implication:** Marketing performance reporting should separate New Customer Acquisition from Retention / Reactivation because their economics are fundamentally different.

## **6. The next budget decision should be based on customer value, not conversion volume**

Campaigns can generate similar conversion volumes while producing different combinations of spend, revenue, ROAS and CPA. Payday Promotion, for example, has a \$61.04 CPA and 1.83x ROAS.

**Business impact:** Conversion volume alone can encourage investment in campaigns that acquire customers at a higher cost.

**Management implication:** Add New Customer CPA, New Customer %, Average Order Value, Gross Margin, 90-day LTV and Repeat Purchase Rate to campaign evaluation.

## **7. Q4 shows a clear deterioration in marketing efficiency**

Monthly ROAS peaks around 1.90–1.92 during parts of May–September, before falling to approximately 1.81 in October and 1.77 in both November and December.

**Business impact:** UrbanCart is generating revenue later in the year, but at a lower return on marketing investment.

**Management implication:** Conduct a Q4 efficiency decomposition covering CPM/CPC inflation, promotional discounting, conversion rate, customer mix, average order value, competitor activity and media saturation.

## **8. Email and SMS are outperforming paid/social channels on engagement and conversion rate**

Visible channel results show Email at 8.62% CTR and 1.81% conversion rate; SMS at 7.58% and 1.57%; Google Ads at 3.44% and 1.12%; Affiliate at 2.83% and 1.00%; Meta Ads at 2.19% and 0.90%; TikTok Ads at 1.81% and 0.69%; Organic Social at 1.22% and 0.49%.

**Business impact:** UrbanCart's owned channels are demonstrating stronger customer response than several acquisition and social channels.

**Management implication:** Maximise customer-database value through segmentation, personalised offers, lifecycle journeys, reactivation campaigns and behaviour-based messaging, validated using incremental revenue.

## **9. Victoria is UrbanCart's largest revenue market, but scale should not be confused with efficiency**

Victoria generates approximately \$65M+ in revenue in the visible state analysis and is the largest market shown.

**Business impact:** Victoria is a major contributor to UrbanCart's overall revenue base.

**Management implication:** Compare marketing spend, conversion, customer acquisition cost, new-customer penetration, customer value and repeat purchase behaviour before changing investment.

## **10. Campaign performance is tightly clustered — so small ROAS differences shouldn't drive major budget moves**

Campaign ROAS in the table is generally clustered around approximately 1.77–1.94.

**Business impact:** There isn't a substantial performance gap between most campaigns based on ROAS alone.

**Management implication:** Avoid major budget reallocations based solely on small observed ROAS differences; assess spend scale, statistical confidence, incremental revenue, customer

acquisition, customer quality, gross margin and repeat purchase behaviour.

## Sensitivity Scenario

If Lead → Conversion improved by just 1 percentage point, from 6.55% to 7.55%, this would represent approximately 331K additional conversions at the current lead volume.

*Note: This is a sensitivity scenario, not a forecast.*

## Customer Conversion Mix

| Customer Type       | Conversions | Share  |
|---------------------|-------------|--------|
| New Customers       | 823K        | 38.26% |
| Returning Customers | 759K        | 35.30% |
| Loyal Customers     | 355K        | 16.51% |
| At-Risk Customers   | 214K        | 9.93%  |

## Management Takeaway

**UrbanCart's core opportunity is not simply to spend more to generate more revenue. The larger opportunity is to improve the quality and efficiency of growth by strengthening lead-to-conversion performance, separating acquisition from retention economics, validating channel incrementality, and evaluating campaigns on customer value rather than conversion volume alone.**

# Power BI Dashboard

The Power BI dashboard is structured around four analytical views.

## 1. UrbanCart Marketing Performance Overview

Business question: How is UrbanCart's overall marketing performance?

**KPIs:** Total Spend • Total Revenue • ROAS • Total Leads • Total Conversions • Conversion Rate

**Analysis / Visuals:** Revenue & Spend Trend • Overall ROAS Trend • Marketing Investment → Business Outcome • Performance vs Previous Period • Monthly Conversion Rate

## 2. Marketing Funnel

Business question: How effectively are users progressing through the marketing funnel?

**KPIs:** Impressions → Clicks → Leads → Conversions

**Analysis / Visuals:** Funnel volume • CTR • Lead Rate • Lead-to-Conversion Rate • Funnel drop-off • Funnel volume trend • Conversion and conversion-rate trend

## 3. Campaign Performance

Business question: How have individual campaigns performed financially and operationally?

**KPIs:** Total Spend • Total Revenue • ROAS • Total Conversions • CPA

**Analysis / Visuals:** Revenue, Cost and ROI by Campaign • ROAS by Campaign • Campaign Performance table • Conversions by Campaign

## 4. Channel & Customer Analysis

Business question: Which channels, customer groups, devices and locations are driving engagement and conversions?

**Analysis / Visuals:** CTR by Channel • Conversion Rate by Channel • Leads & Conversions by Channel • Conversions by Customer Segment • Conversion Rate by Device • Conversions by Geography

## Full Campaign Performance Results

| Metric            | Result           |
|-------------------|------------------|
| Total Spend       | <b>\$130.97M</b> |
| Total Revenue     | <b>\$241.36M</b> |
| ROAS              | <b>1.84x</b>     |
| Total Conversions | <b>2,168,828</b> |
| CPA               | <b>\$60.39</b>   |
| ROI               | <b>84.29%</b>    |

These represent the full Campaign Performance view and may differ from values displayed on filtered dashboard pages.

# Final Recommendations

## 1. Investigate high-efficiency campaigns

New Product Launch and EOFY Deals showed strong historical ROAS and relatively low CPA. Their audience, offer, channel mix, timing and creative could be investigated to identify transferable practices.

## 2. Analyse high-volume campaigns separately

Payday Promotion, Tech Upgrade, Refer a Friend and Black Friday generated high conversion volumes. These campaigns can be evaluated for their ability to generate scale as well as efficiency.

## 3. Investigate lower-efficiency campaigns

VIP Customer Offer could be investigated further to understand whether audience, offer, channel mix, timing or acquisition cost contributed to its lower ROAS.

## 4. Use historical results as benchmarks

Historical ROAS, CPA and conversion performance can be used as benchmarks for future campaign tests. Historical performance should be treated as evidence for testing, not as a guarantee of future performance.

# How to Run This Project

- Clone the repository.
- Open the .pbix file in Power BI Desktop.
- Update the data source path if required.
- Refresh the dataset.
- Explore the dashboard.

# Skills Demonstrated

- Data cleaning
- Data quality analysis
- Data transformation
- Data modelling
- Power Query
- DAX
- KPI development
- Marketing analytics
- Funnel analysis
- Campaign performance analysis
- Customer segmentation
- Dashboard design
- Business storytelling

- Insight generation
- Stakeholder-focused reporting

## Disclaimer

UrbanCart is a fictional company created for portfolio and learning purposes. The dataset is synthetic and does not represent real UrbanCart business performance.

The recommendations are analytical suggestions based on the available historical data and should be validated with additional business, financial and customer information before being used for real-world decisions.

## Author

**Tejaswi Aluka**

Data Analyst / BI Portfolio Project

Email: [alukatejaswi@gmail.com](mailto:alukatejaswi@gmail.com)

LinkedIn: [linkedin.com/in/tejaswi-aluka-9240a3403](https://www.linkedin.com/in/tejaswi-aluka-9240a3403)

Skills: SQL | Power BI | Excel | Python | DAX